Where to Start Cutting Chicago's Building Emissions

Chicago aims to cut its emissions 62% by 2040 from 2017 levels (City of Chicago, 2022). Buildings make up a majority of emissions in Chicago. In 2013, the City released the Chicago Energy Benchmarking Ordinance which requires large buildings greater than 50,000 square feet to report yearly data on their emissions. This analysis uses Chicago Energy Benchmarking data from 2014–2023 to examine trends in building emissions and to identify targets for impactful emissions reduction.

Important terms

CO2e
Carbon dioxide equivalent. It converts every greenhouse gas to the amount of carbon dioxide that would trap the same heat, so emissions can be added up in one unit.

Ton
A metric ton or 1,000 kilograms. One ton of CO2e is equivalent to what a gasoline car emits in under three months or driving about 2,500 miles (U.S. Environmental Protection Agency [EPA], 2026).

Emissions vs. energy use
Emissions come from how much energy a building uses and how clean that energy is. Both using less energy and switching to cleaner energy lowers emissions.

Emission factor
How much CO2e a fuel produces for each unit of energy, in kilograms of CO2e per kBtu (1 kWh is about 3.4 kBtu). Emissions = energy used x emission factor. The data does not list the factors, so we estimated them from the buildings' fuel use and reported emissions. Figure 3 uses 2018 factors from its 745 buildings (electricity 0.167, natural gas 0.052). Figure 4 uses 2023 factors from all reporting buildings (electricity 0.139, natural gas 0.054).

Emissions intensity or emissions per square foot
A building's yearly emissions divided by its floor area, in kilograms of CO2e. This allows for comparing large and small buildings.

Reporting buildings
Buildings that submitted benchmarking data. Up to 2,584 reported for 2023, and each chart uses those with the data it needs.

Chicago Energy Rating
A rating of 0 to 4 stars. 4 is the best, 1 is the worst. Zero stars are given to buildings that did not submit a complete report. For rating-related analysis, these are excluded and not counted as poorly rated.

ENERGY STAR score
A score from 1 to 100 that compares a building's energy use with similar buildings nationwide. The Chicago Energy Rating is based on it.

Percentile (within a property type)
The share of buildings of the same type that a building emits more per square foot than. A building at the 77th percentile emits more than 77% of its peers. *Top quarter* means the 75th percentile or higher.

Acronyms
CMAP: Chicago Metropolitan Agency for Planning. EPA: U.S. Environmental Protection Agency.

Background

Buildings consistently produce about two-thirds of the city's emissions: 62% in CMAP's 2019 inventory (17 of 27.5 million tons) and 70% in the City's 2025 benchmarking release (Chicago Metropolitan Agency for Planning [CMAP], 2022; City of Chicago, Office of the Mayor, 2025).

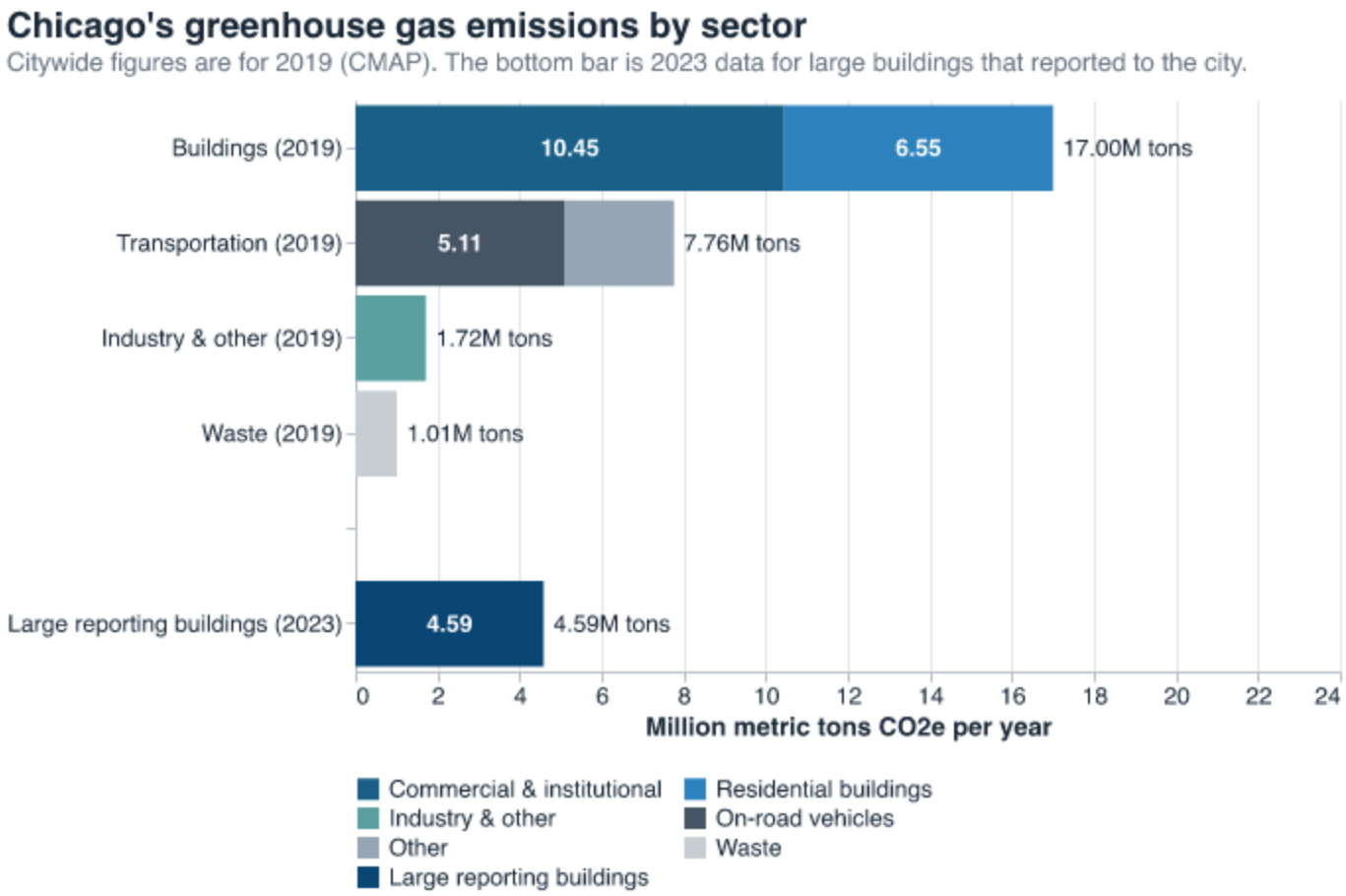


Figure 1. Buildings are Chicago's biggest source of emissions. Source: Chicago Metropolitan Agency for Planning, Regional Greenhouse Gas Emissions Inventory (2019 data); Chicago Energy Benchmarking (2023).

The 2,584 buildings that reported in 2023 emitted 4.6 million tons, about a quarter of the 17 million tons from buildings in CMAP's 2019 inventory (a different year and method). Note that a portion of the missing tons come from all buildings under 50,000 square feet that aren't included in this analysis. Another 542 required buildings did not report, so about 83% of the 3,126 required reporters submitted emissions data. Another 312 buildings had an undefined status and are not included in this analysis.

Current state of reporting buildings

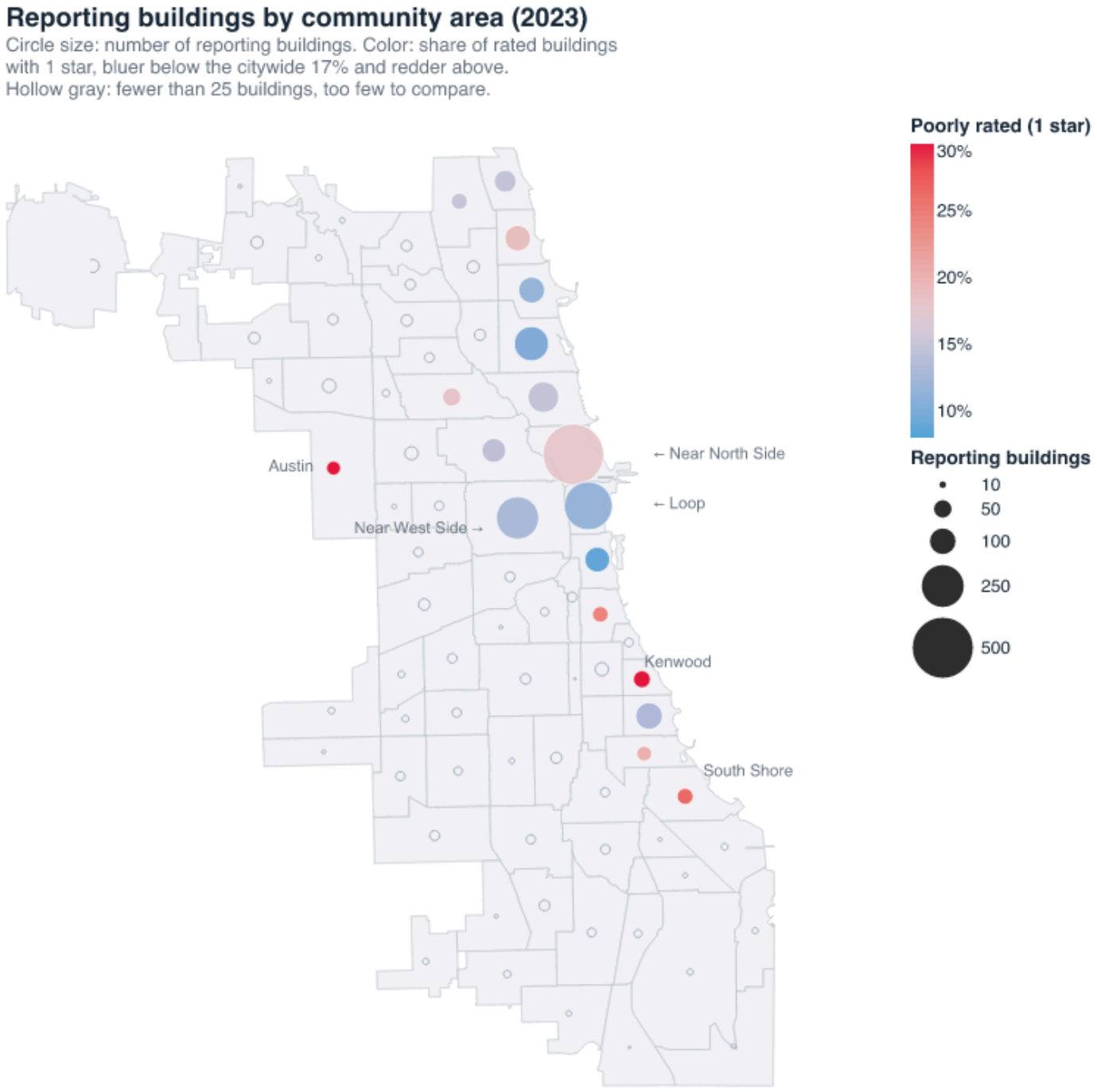


Figure 2. Buildings cluster downtown, but the highest shares of poorly rated buildings are elsewhere. Source: Chicago Energy Benchmarking (2023). The map includes buildings over 50,000 square feet only.

A high-level overview of Chicago's buildings shows the share of buildings is concentrated downtown, in the Loop and the Near North, Near West and Near South Sides. Though Downtown has 42% of reporting buildings, only 14% of its rated buildings are poorly rated. The highest shares of poorly rated buildings are in Kenwood (34%), Austin (31%) and South Shore (26%), though each has only 26 to 38 buildings. Stars come

from the Chicago Energy Rating (4 is best, 1 is worst) and are based on a building's ENERGY STAR score (City of Chicago, n.d.). Ratings alone do not show how much a building emits, so the next figures turn to emissions.

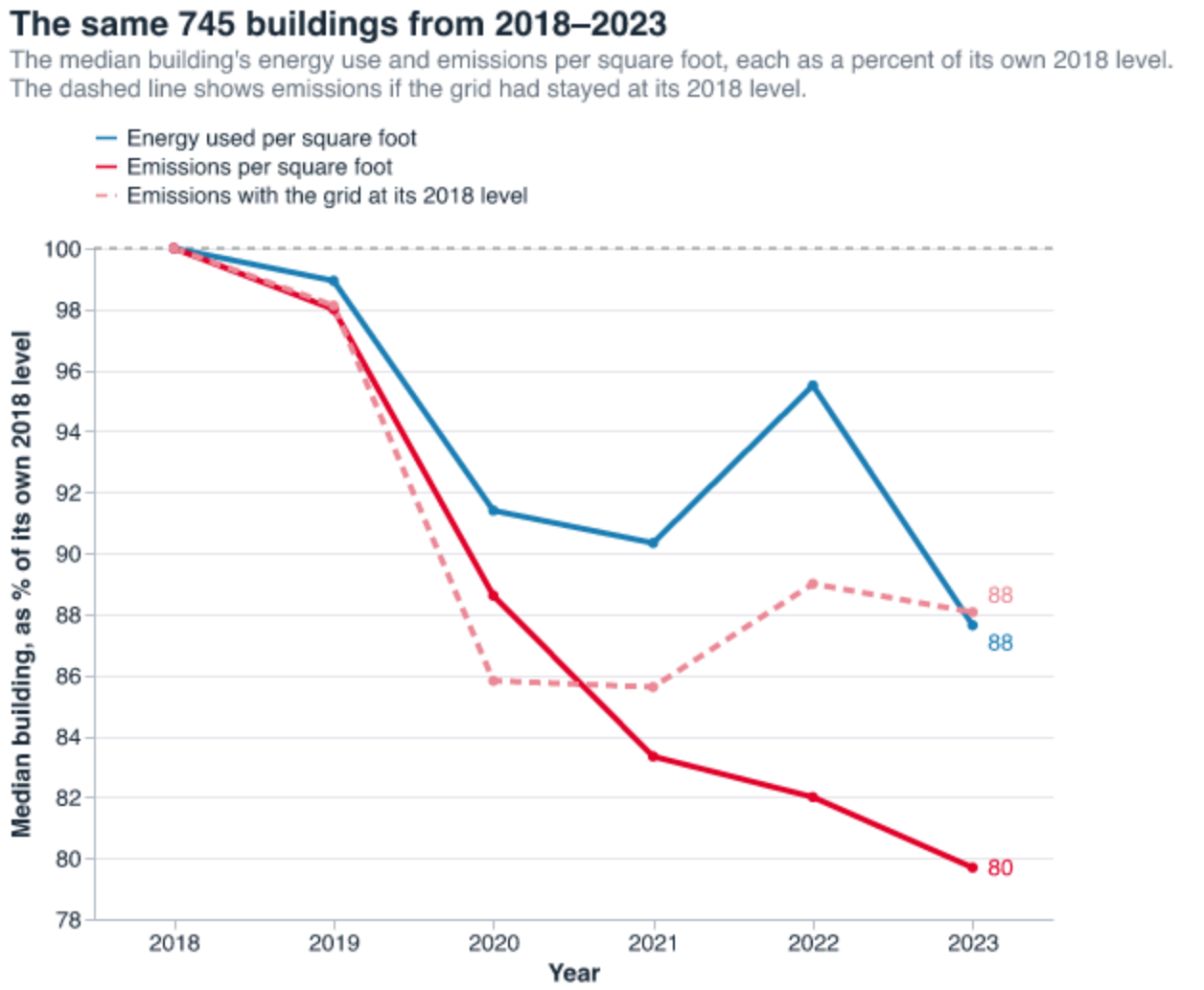


Figure 3. Emissions per square foot fell 20% since 2018, but energy use fell only 12%. The pale dashed line shows emissions per square foot if the grid had stayed at its 2018 level. Source: Chicago Energy Benchmarking (2018 to 2023).

For the 745 buildings that reported every year, energy use fell 12% since 2018 but emissions fell 20%. The dashed line converts each year's fuel use to emissions with the 2018 emission factors (kilograms of CO2e per unit of electricity, gas and other fuels), which we estimated from these buildings' own 2018 fuel use and reported emissions. With the factors held fixed, emissions follow energy use and end at 88, the same as energy use. The 8-point gap down to the solid red line (80) is the effect of electricity becoming cleaner: the same energy produces less pollution.

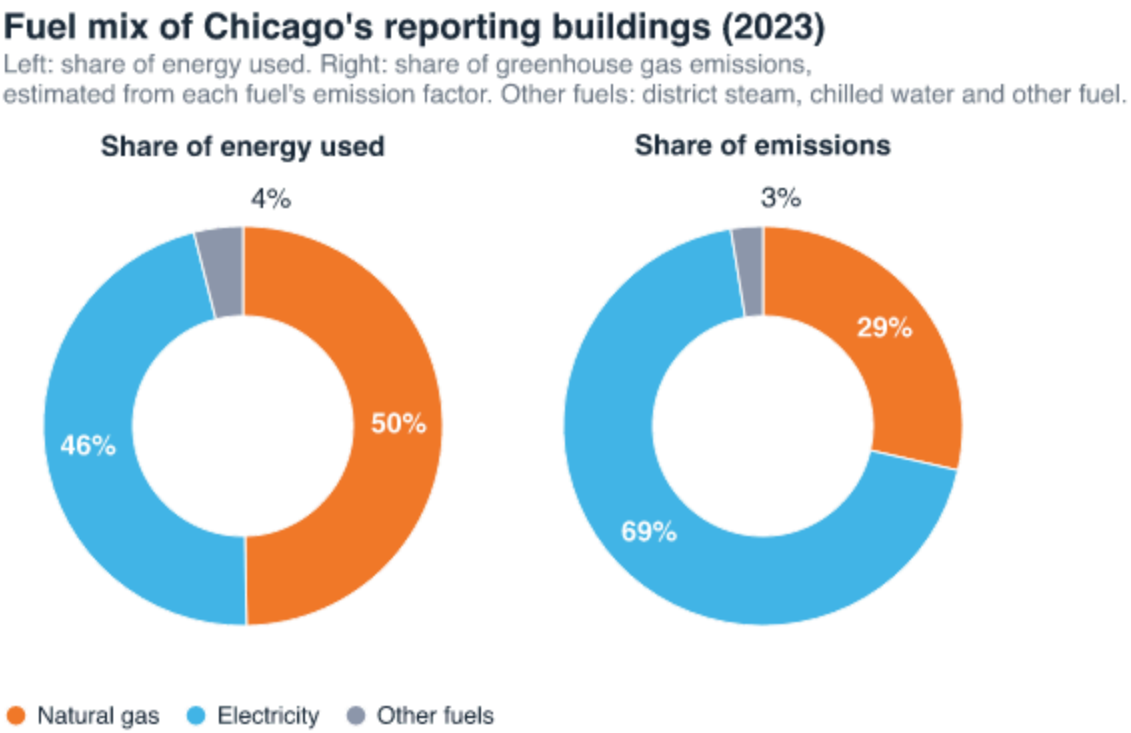


Figure 4. Natural gas supplies about half of the energy used, but electricity produces about 70% of the emissions. Left: each fuel's share of the energy used. Right: each fuel's share of the emissions, estimated by multiplying the energy each fuel supplied by how much pollution it produces per unit of energy (its emission factor). Electricity produces more per unit than gas because of how the grid generates power. Other fuels: district steam, chilled water and other fuel. Source: Chicago Energy Benchmarking (2023).

Gas (50%) and electricity (46%) supply nearly all of the energy these buildings use. They are not equally polluting: a unit of electricity produces more emissions than a unit of gas, so electricity accounts for about 69% of the emissions and gas for 29%. That share depends on the grid, so as electricity gets cleaner its share of emissions falls, and replacing gas heating with electric heat can become more worthwhile.

Emissions and emissions intensity by building type

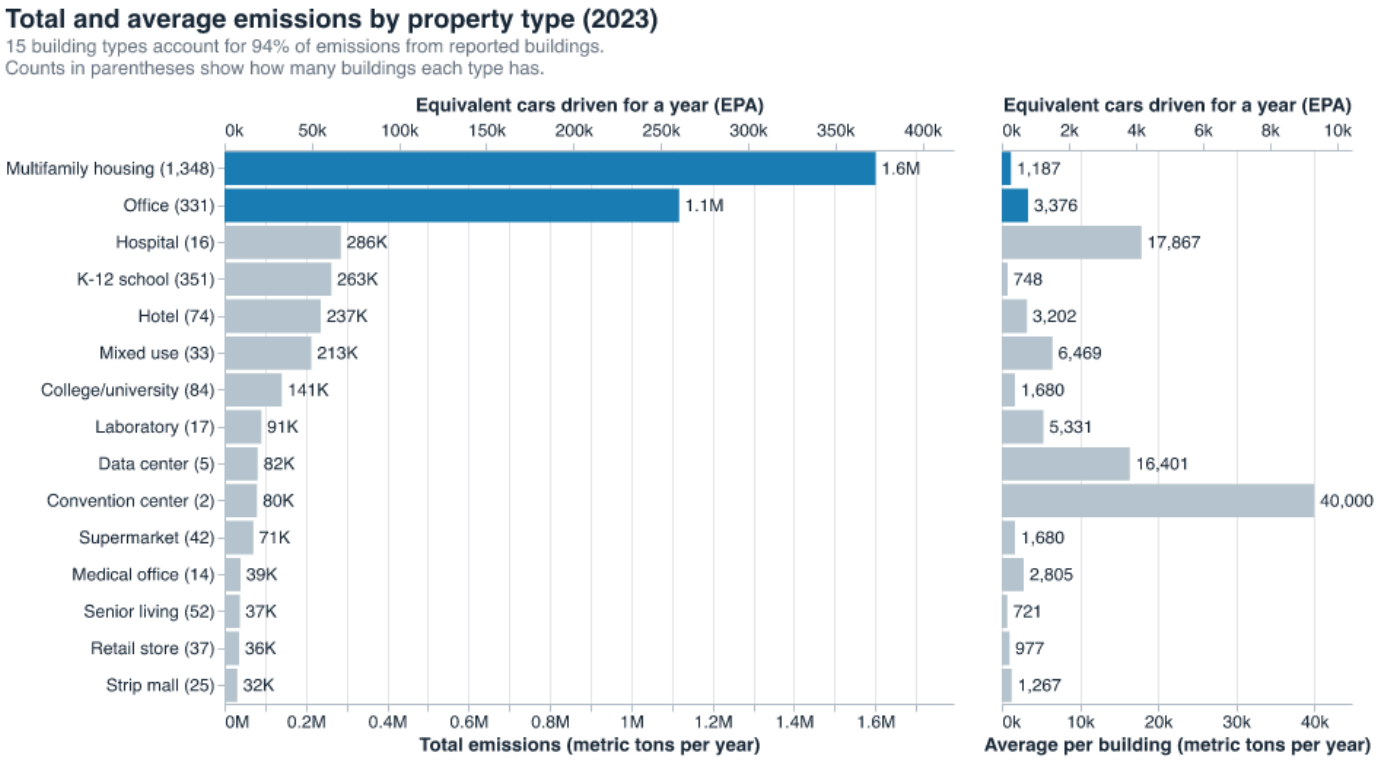


Figure 5. Multifamily housing and offices emit the most in total, but each multifamily building emits little. Left: total emissions. Right: the average emissions of one building of that type. Source: Chicago Energy Benchmarking (2023).

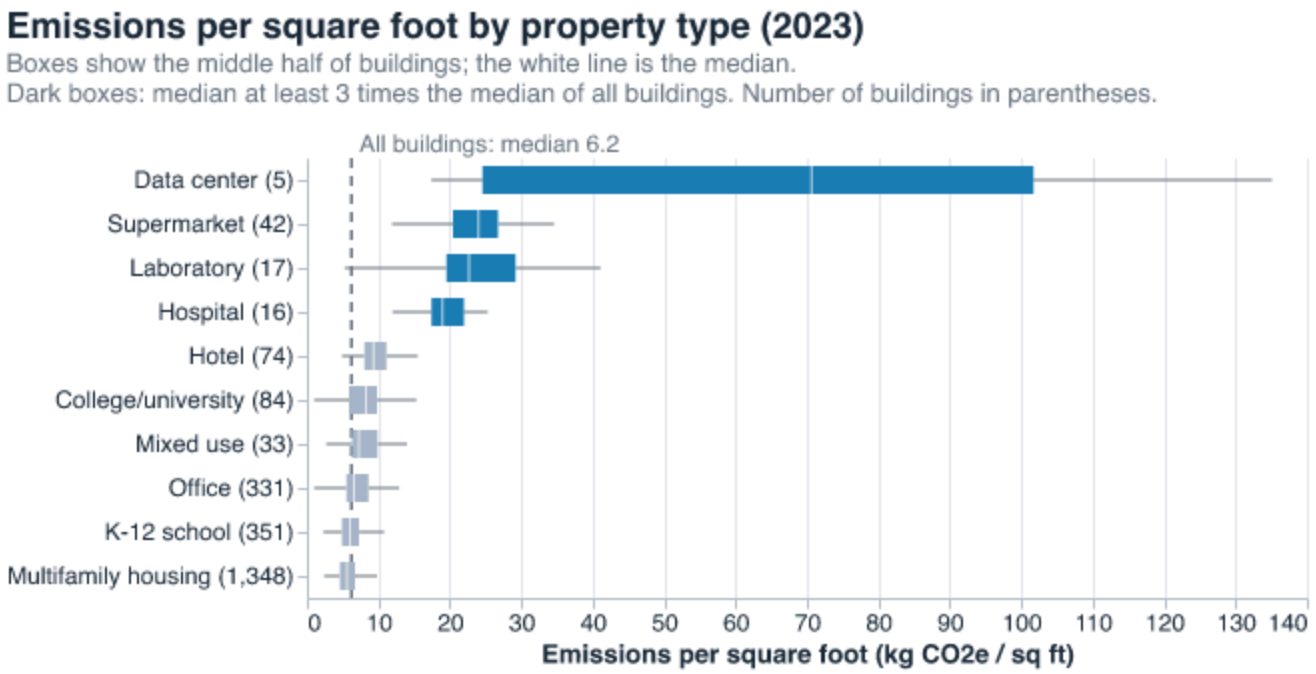


Figure 6. Size and intensity are different problems. Ten property types: the seven with the most total emissions plus any type with at least 5 buildings whose median is at least 3 times the median of all buildings. Boxes show the middle half of buildings, the white line is the median, and the thin lines reach the farthest building within 1.5 box lengths of the box. Dark boxes are types whose median is at least 3 times the median of all buildings, the dashed line at 6.2. Outliers are not shown. Source: Chicago Energy Benchmarking (2023).

Multifamily housing and offices were the property types with the most emissions. However, they also have the top and third top building counts, 1,348 for multifamily housing and 331 for offices. When you look at the average emissions, convention centers, hospitals and data centers emit the most per building. The average hospital emits about 15 times as much as the average multifamily building. Together, multifamily housing and offices produce 59% of reported emissions from many modest buildings, so reaching many ordinary buildings matters. Hospitals, laboratories, data centers and supermarkets, which are 3 to 12 times as intense as the typical building, produce about 12% from far fewer buildings, so they call for targeted, building-by-building work.

Buildings to target for emissions reductions

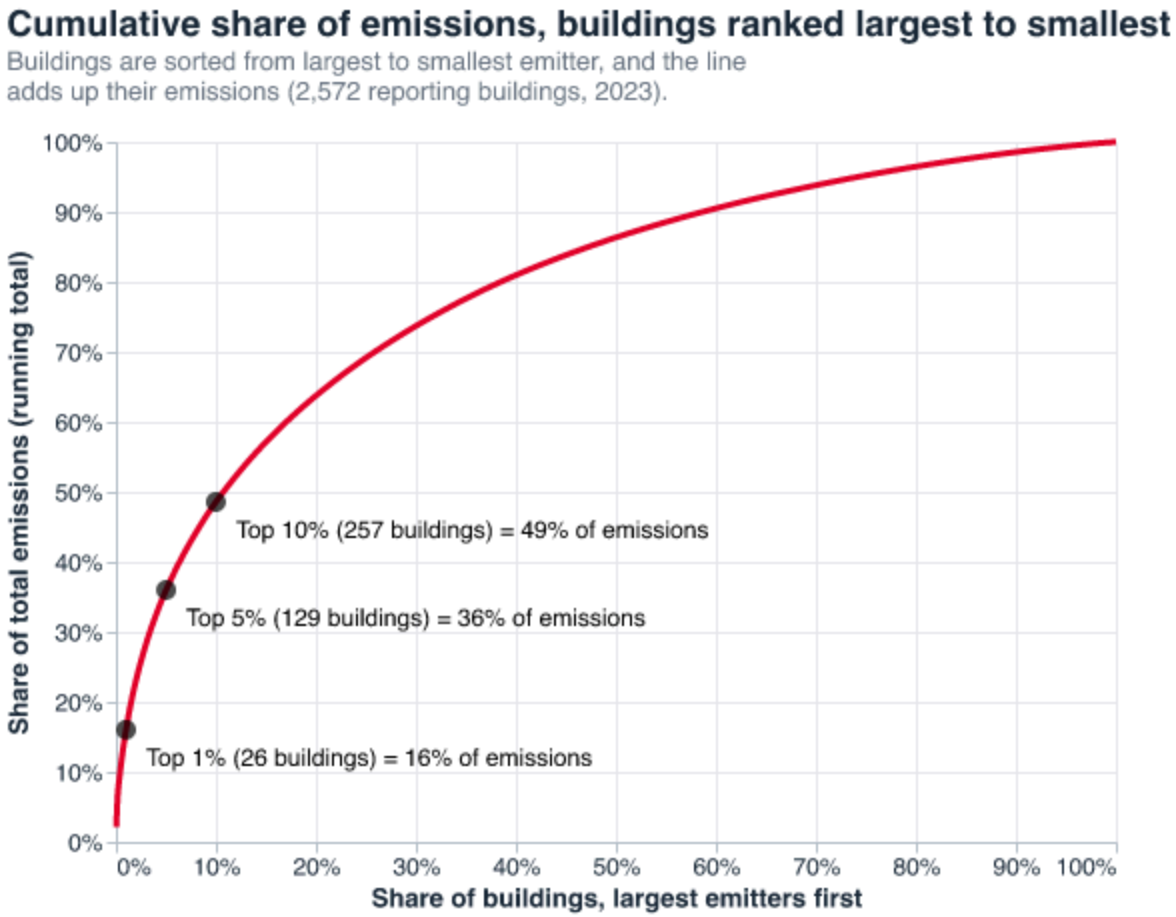


Figure 7. Half of reported emissions come from 10% of buildings. Source: Chicago Energy Benchmarking (2023).

The 26 largest buildings (1%) produce 16% of reported emissions, the largest 129 (5%) produce 36%, and the largest 257 (10%) produce 49%. A few hundred buildings carry about half of the emissions, so the next figure asks which of them are also poorly rated.

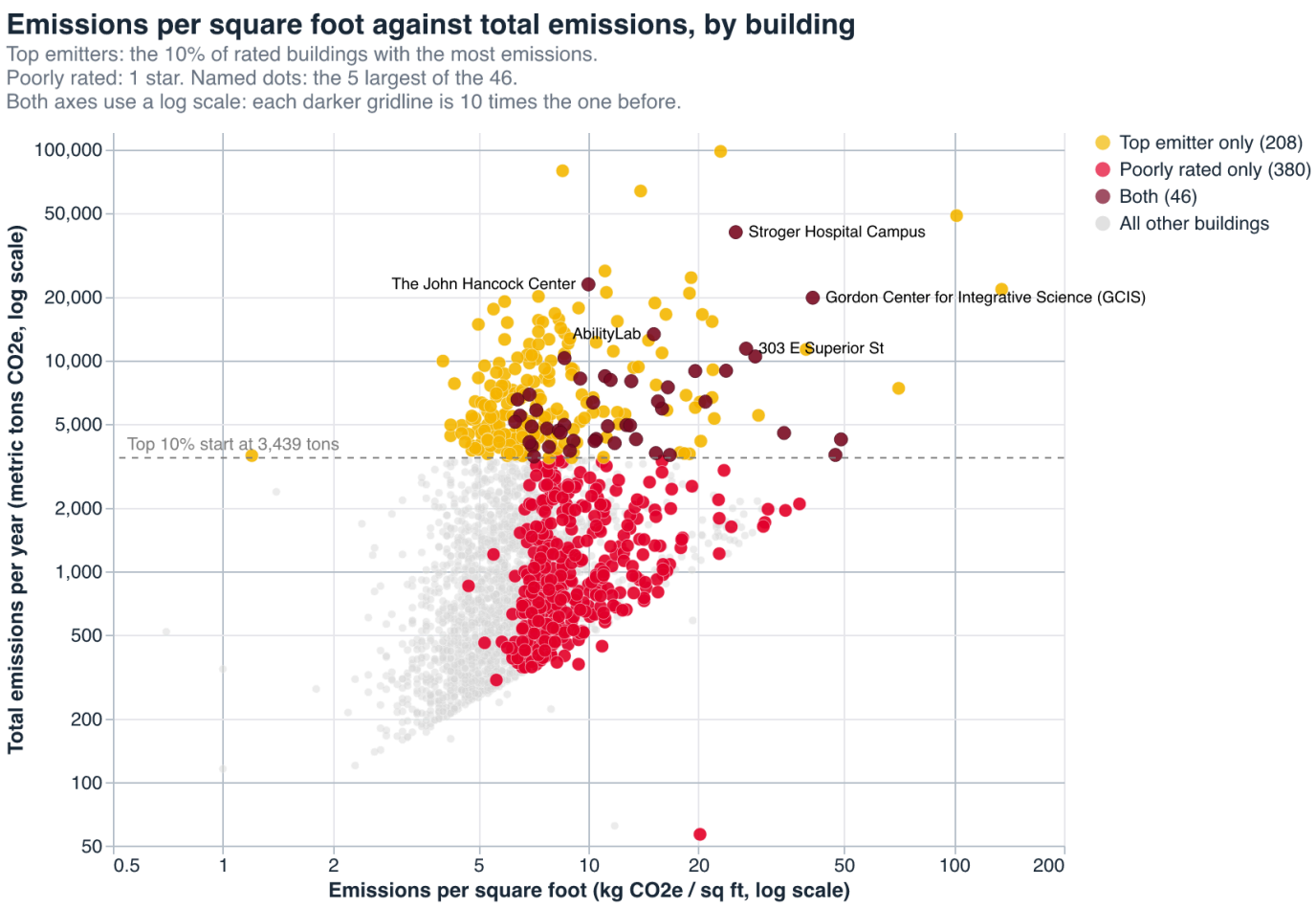


Figure 8. 46 big buildings are poorly rated; 37 could cut about 87,000 tons. Source: Chicago Energy Benchmarking (2023).

A poor rating reflects real waste, but it is not where most of the emissions are. The median 1-star building emits 8.4 kg per square foot, against 4.6 for 4 stars, yet about 9 in 10 poorly rated buildings are modest emitters. Only 46 are also among the largest, and together they produce 338,000 tons, 7% of reported emissions. Stroger Hospital Campus (about 40,000 tons), the John Hancock Center (22,800) and the Gordon Center for Integrative Science (19,700) lead the list. 37 of the 46 are in the top quarter of their type for emissions per square foot, and if they matched the typical building of their type, emissions would fall about 87,000 tons a year, roughly 2% of reported emissions (City of Chicago, 2025).

Data Sources

- **Chicago Energy Benchmarking** (City of Chicago Data Portal): annual energy use, greenhouse gas emissions and Chicago Energy Rating for buildings over 50,000 square feet. This analysis uses 2023 for most charts and 2018 to 2023 for the trend. Data last updated by the city February 5, 2025; downloaded October 1, 2026. https://data.cityofchicago.org/Environment-Sustainable-Development/Chicago-Energy-Benchmarking/xq83-jr8c
- **CMAP Regional Greenhouse Gas Emissions Inventory** (2022 report, 2019 data): citywide emissions by sector. It is the most recent citywide inventory available, and it comes from a different year and method than the benchmarking data.
- **Boundaries - Community Areas** (City of Chicago Data Portal): the community area shapes for the map (City of Chicago, 2026). Last updated by the city March 25, 2026. https://data.cityofchicago.org/resource/igwz-8jzy

Sources

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